Coefficient of self inductance for a long selenoid

N = total no of turns

l = length of solinoid

R = radius of each coil

n = number of term per unit length

$$= \frac{N}{l}$$

B = magnetic field within the solenoid

$$= \mu_0 i n$$

ϕ linked with each coil $= BA = \mu_0 i n \pi R^2$

Total flux $\phi_T = N\phi = N\mu_0 n i \pi R^2$

$$= (nl)\mu_0 n i \pi R^2$$

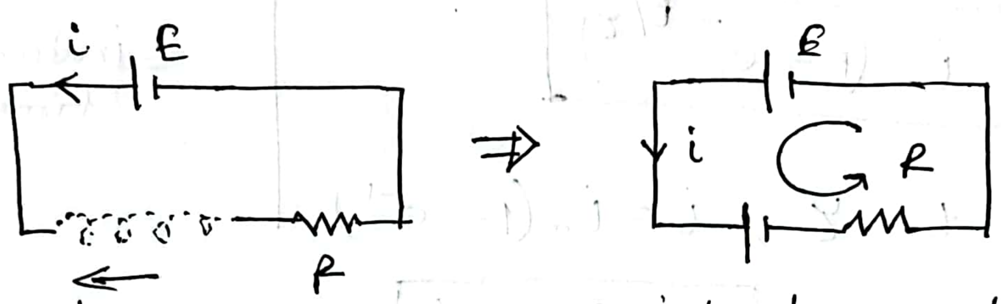
$$= \mu_0 n^2 \pi R^2 l\, i$$

$$\Rightarrow L i = \mu_0 n^2 \pi R^2 l\, i$$

$$\Rightarrow \boxed{L = \mu_0 n^2 \pi R^2 l}$$

Coeft. of self inductance depends on geometry.

Certain time $t = t'$ where $0 < t' < \infty$, $i = i'$ and $0 < i < i_0$



induced current

as inductor produces EMF we can draw a battery

Now, applying Kirchhoff's law :

$$E - L\frac{di}{dt} - iR = 0$$

$$\Rightarrow E - iR = L\frac{di}{dt}$$

$$\Rightarrow \int_0^t \frac{dt}{L} = \int_0^i \frac{di}{E - iR}$$

$$\Rightarrow \frac{t}{L} = -\frac{1}{R}\log_e\left[E - iR\right]_0^i$$

$$\Rightarrow -\frac{Rt}{L} = \log_e|E - iR| - \log_e|E|$$

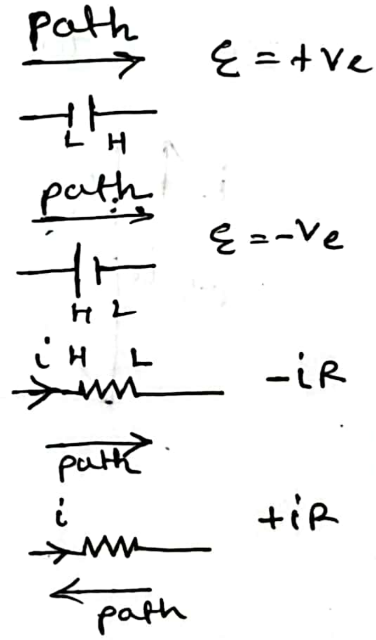
$$\Rightarrow -\frac{Rt}{L} = \log_e\left|\frac{E - iR}{E}\right|$$

$$\Rightarrow -\frac{Rt}{L} = \log_e\left|1 - \frac{iR}{E}\right|$$

$$\Rightarrow 1 - \frac{iR}{E} = e^{-(R/L)t}$$

$$\Rightarrow \frac{iR}{E} = 1 - e^{-(R/L)t}$$

$$\Rightarrow i = \frac{E}{R}\left(1 - e^{-(R/L)t}\right)$$



$$\int \frac{dx}{b - ax} = \frac{\log_e|b - ax|}{-a}$$

$$\Rightarrow i = i_0 \left(1 - \cancel{e}\, e^{-(R/L)t}\right)$$

$$\boxed{i = i_0 \left(1 - e^{-t/\tau}\right)}$$

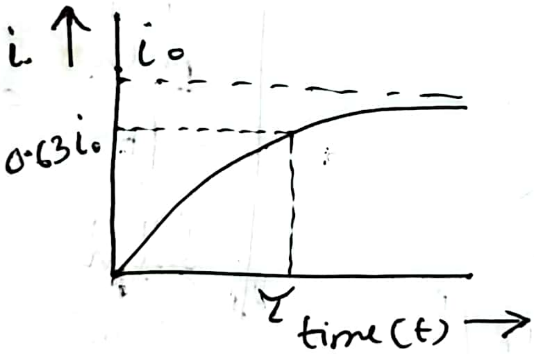
$\tau = \frac{L}{R}$

= inductive time const

When, $t = \tau$; $i = i_0(1 - e^{-1})$

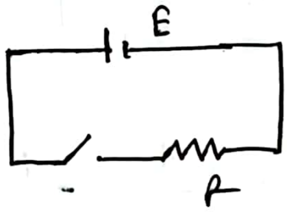
$$\Rightarrow \boxed{i = 0.63\, i_0}$$

τ is that time in which current grows from zero to 63% of its maximum value.



at $t = 0$, $i = 0$

No current flows in branch containing inductor

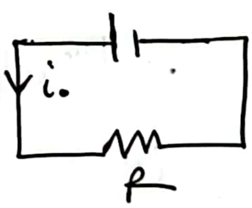


open circuit inductor

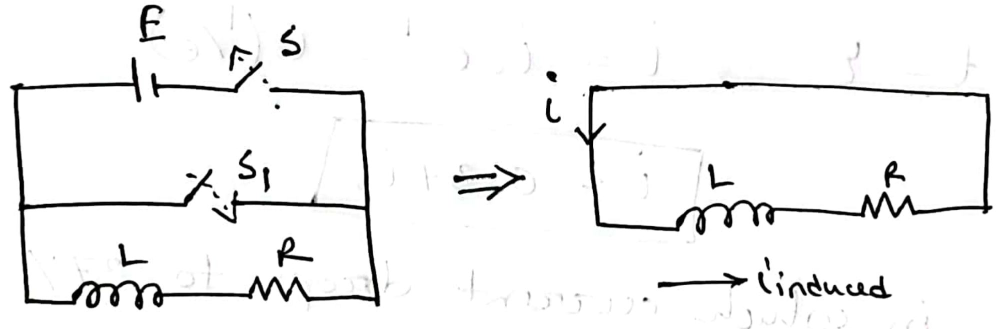
at $t = \alpha$, steady state $i = i_0$

$i = i_0\,(1 - e^{-\alpha})$

$= i_0 \left(1 - \frac{1}{e^{\alpha}}\right) = i_0\,(1 - 0) = i_0$



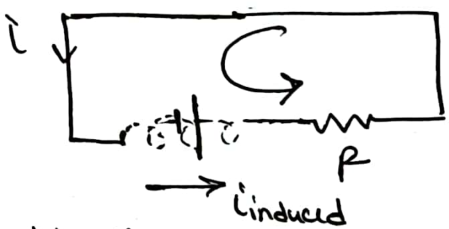
Short circuit inductor.



After closing S. current reaches its max value $i_0 = \frac{E}{R}$

Then we open 'S' and close 'S_1'

- As S is opend, battery removed, so the current should be instantly zero, but it will not be the case due to L
- Inductor will not let current to ~~dy~~ die as inductor opposes any change.
- Induced current in the same direction.



Appy Kirchhoff's law,

$$-L\frac{di}{dt} - iR = 0$$

[as current decreasing so $\frac{di}{dt}$ is -ve

$$\Rightarrow -iR = L\frac{di}{dt}$$

At $t \to$
$i = 0$
$t = 0,\ i = i_0$
decreasing from max

$$\Rightarrow -\frac{R}{L}\int_0^t dt = \int_{i_0}^{i} \frac{di}{i}$$

$$\Rightarrow -\frac{Rt}{L} = \log_e i - \log_e i_0$$

$$\Rightarrow -\frac{Rt}{L} = \log_e\left(\frac{i}{i_0}\right)$$

$$\Rightarrow \frac{i}{i_0} = e^{-(R/L)t}$$

$$\Rightarrow \boxed{i = i_0 e^{-t/\tau}}$$

At, $t = \tau$, $i = i_o e^{-1} = i_o(1/e)$

$$\boxed{i = 0.37\, i_o}$$

Time in which current drops to 37% of initial.